

Biomarkers in Treating Depression with Ayahuasca

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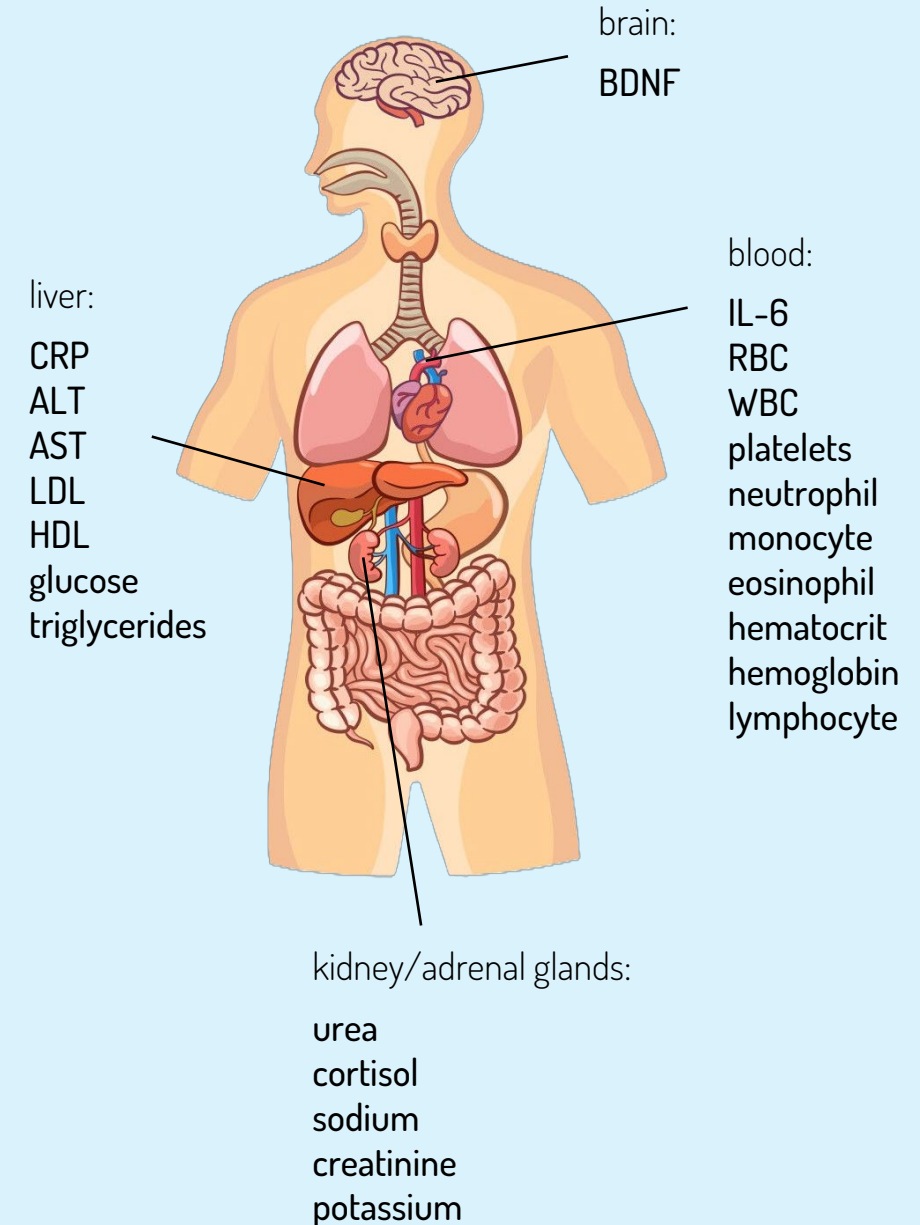
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Ayahuasca clinical trial for depression

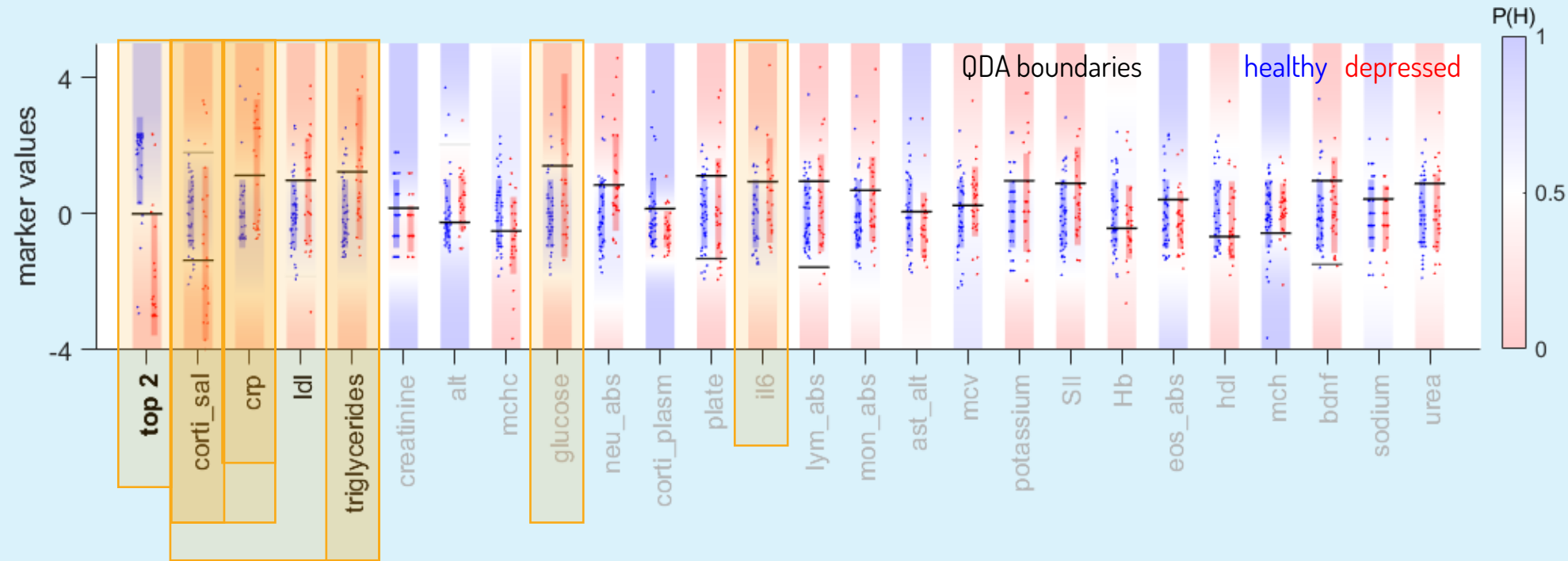
- 46 healthy, 29 with treatment-resistant depression
- Each group split into **ayahuasca** and **placebo**
- 25 biomarkers collected before and after treatment

Questions

- How do biomarkers differ between healthy and depressed people?
- How do biomarkers change after treatment in healthy and depressed people?
- Which biomarker changes correlate with reductions in depression?

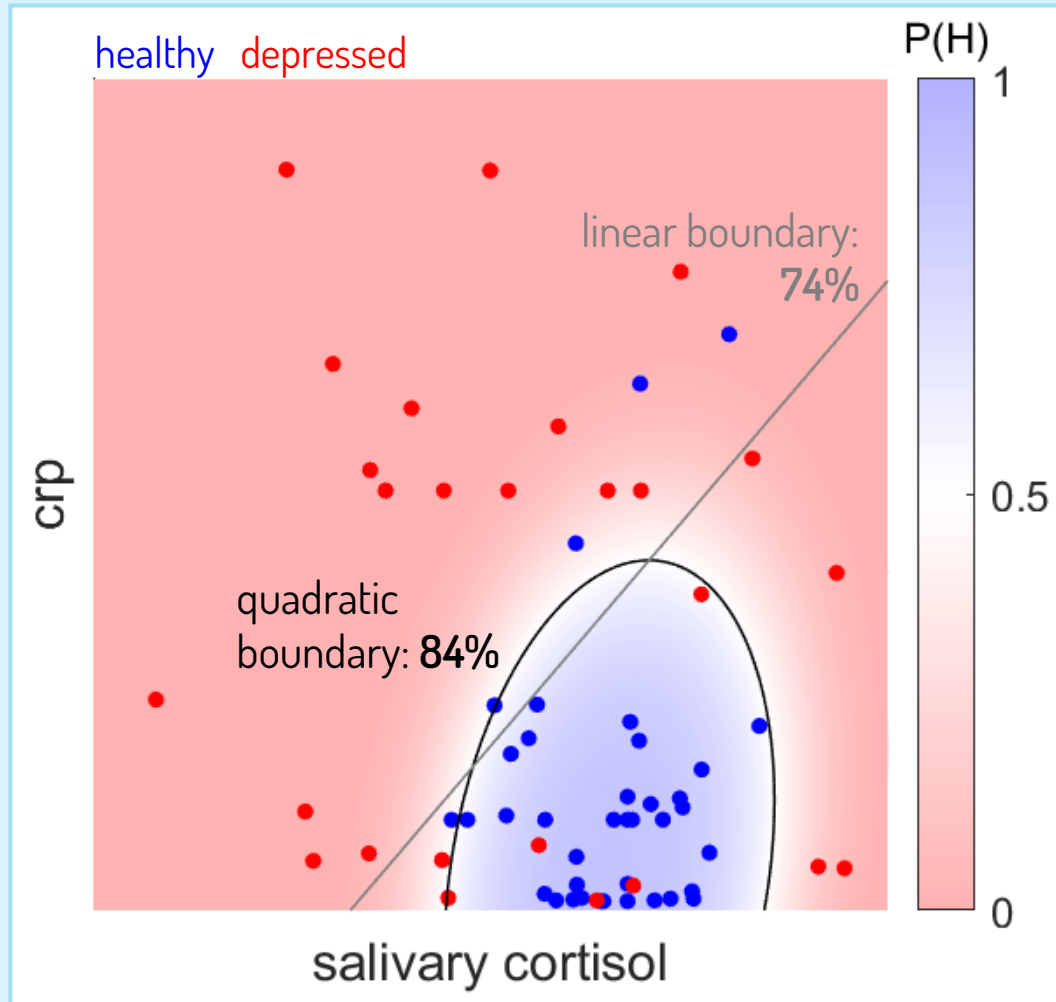


1. How do biomarkers differ between healthy and depressed?



- Depressed have **both low and high salivary cortisol**.
Low: fatigue / High: stress
- Here we need **2 boundaries**.
- In depressed, CRP, triglycerides, glucose & IL-6 are higher.
- 1 boundary is good enough for most markers.
- Only **4 markers** have significant **cross-validated** separation.
- Salivary cortisol** and **CRP** are the **best individual markers**: **75% separation** each.
- CRP: inflammation, fatigue, anhedonia.
- Top 2 markers **combined** is even better: **84% separation**.

Combining top 2 markers to separate healthy/depressed



Colour-map is the **Bayesian posterior probability** of healthy.

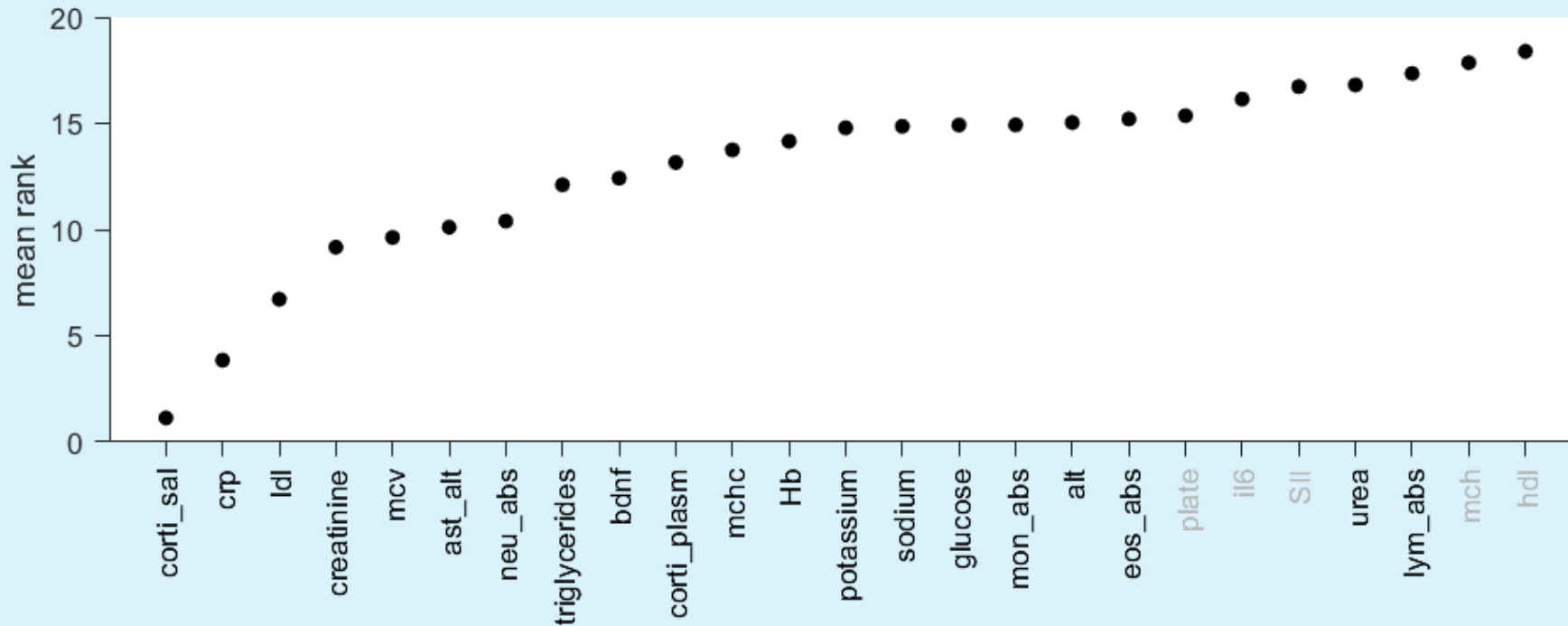
Quadratic boundary exploits narrow range of cortisol in healthy, linear boundary can't.

When combining a new marker, only **new info** counts.

Added CRP not because it is second-best, but because it **adds the most separation**.

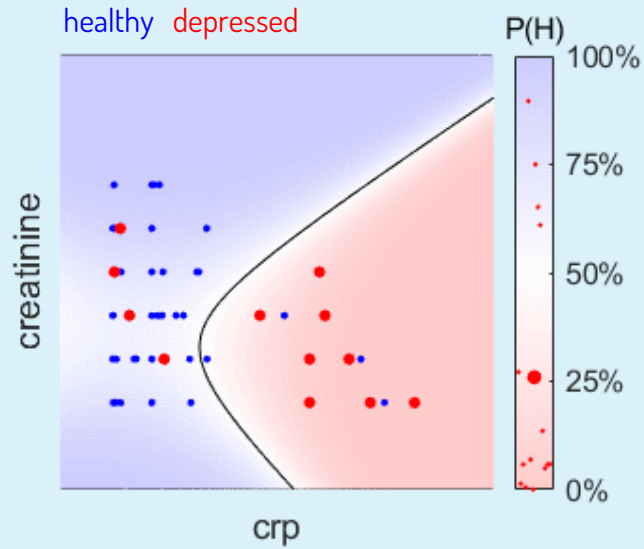
$P(H)$ is the **single combined marker**.

Building an optimal combination of **separation** markers



- Used **greedy algorithm** to add markers one by one.
- Found mean ranks across many runs
- **Combined separation** first rises (**underfitting**), then falls (**overfitting**).
- Quadratic boundaries are significantly better than linear for up to 15 dimensions.
- Combining **salivary cortisol** and **CRP** is good enough
- Beyond these 2, best markers are not in order of individual separation accuracy

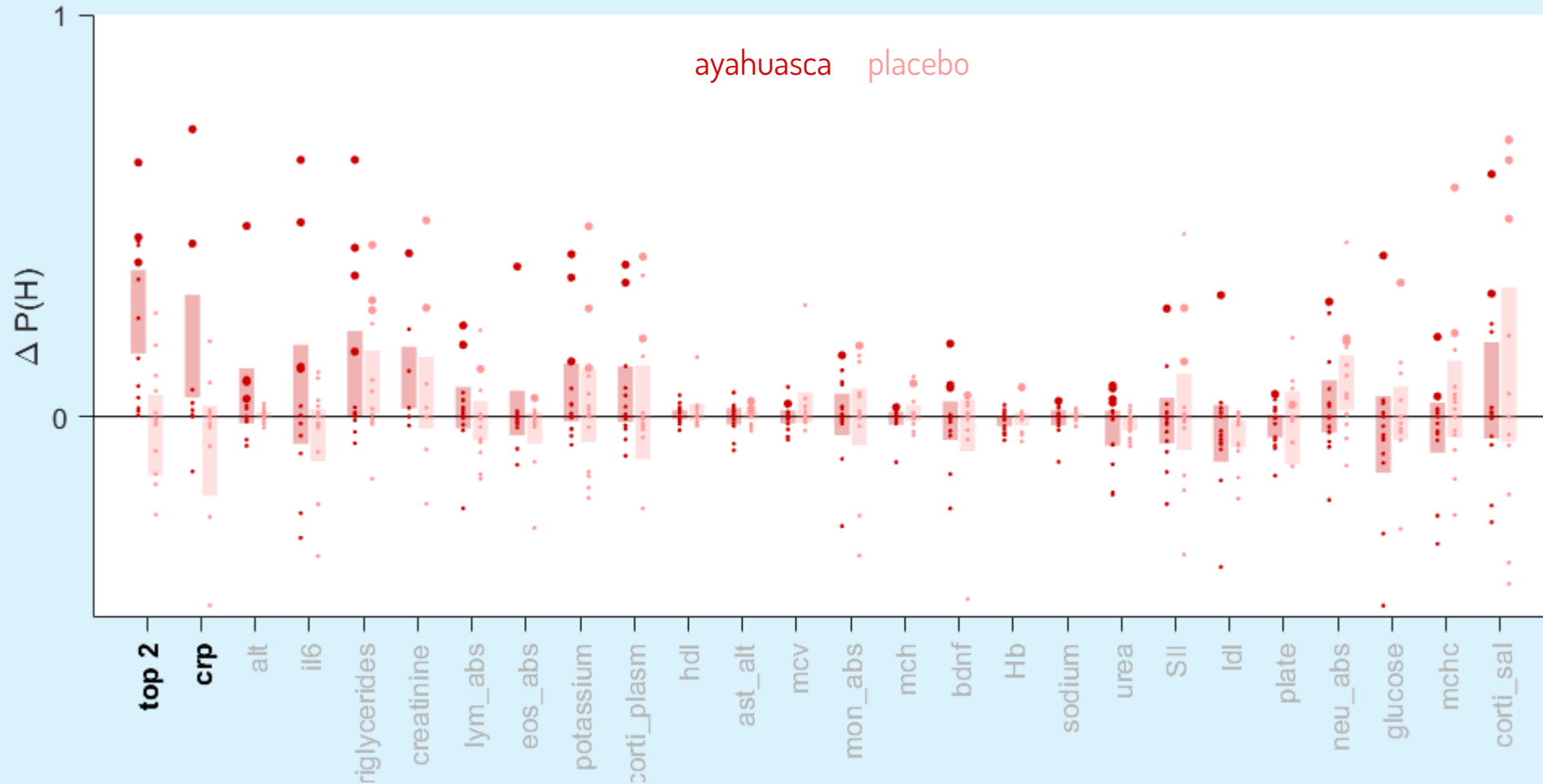
ayahuasca



2. How do markers change after treatment?

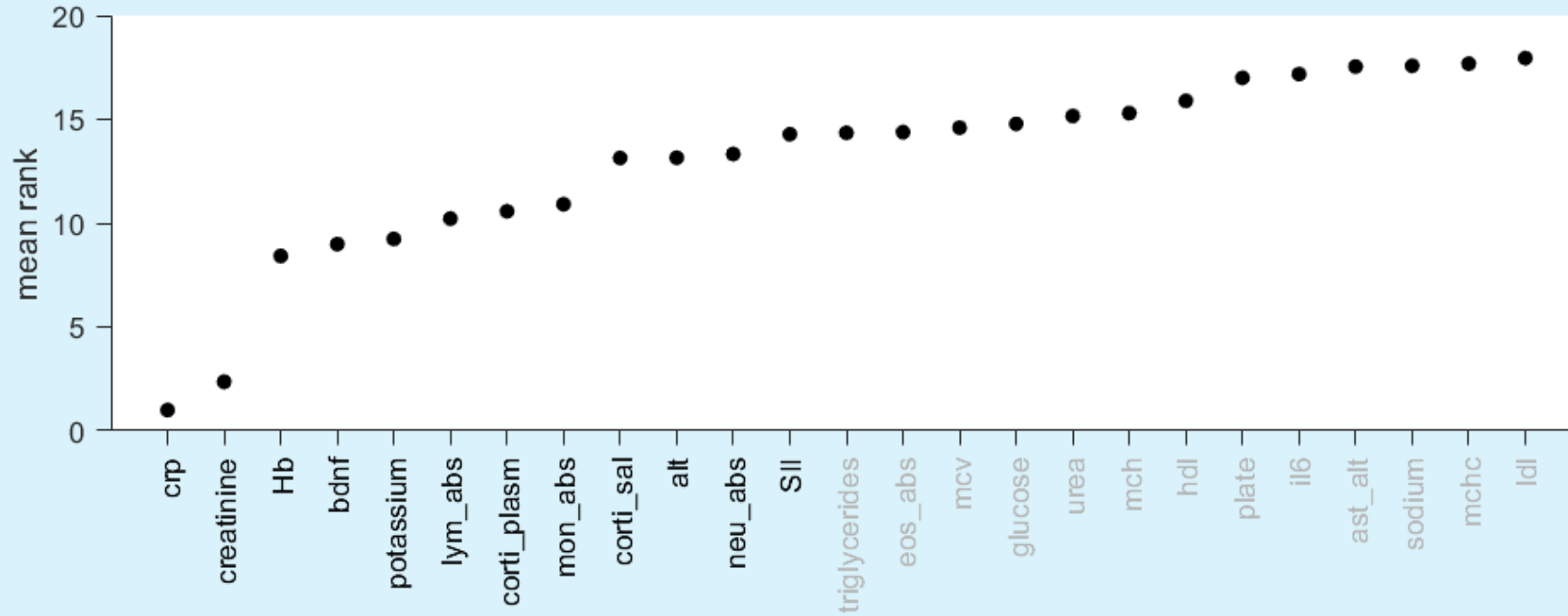
- These markers are being restored towards healthy values after ayahuasca treatment.
- Average $P(H)$ rises from 25% to 50%.
- No such change after placebo.
- No overall change for the healthy, with ayahuasca or placebo.

2. How are markers of the **depressed** restored after treatment?



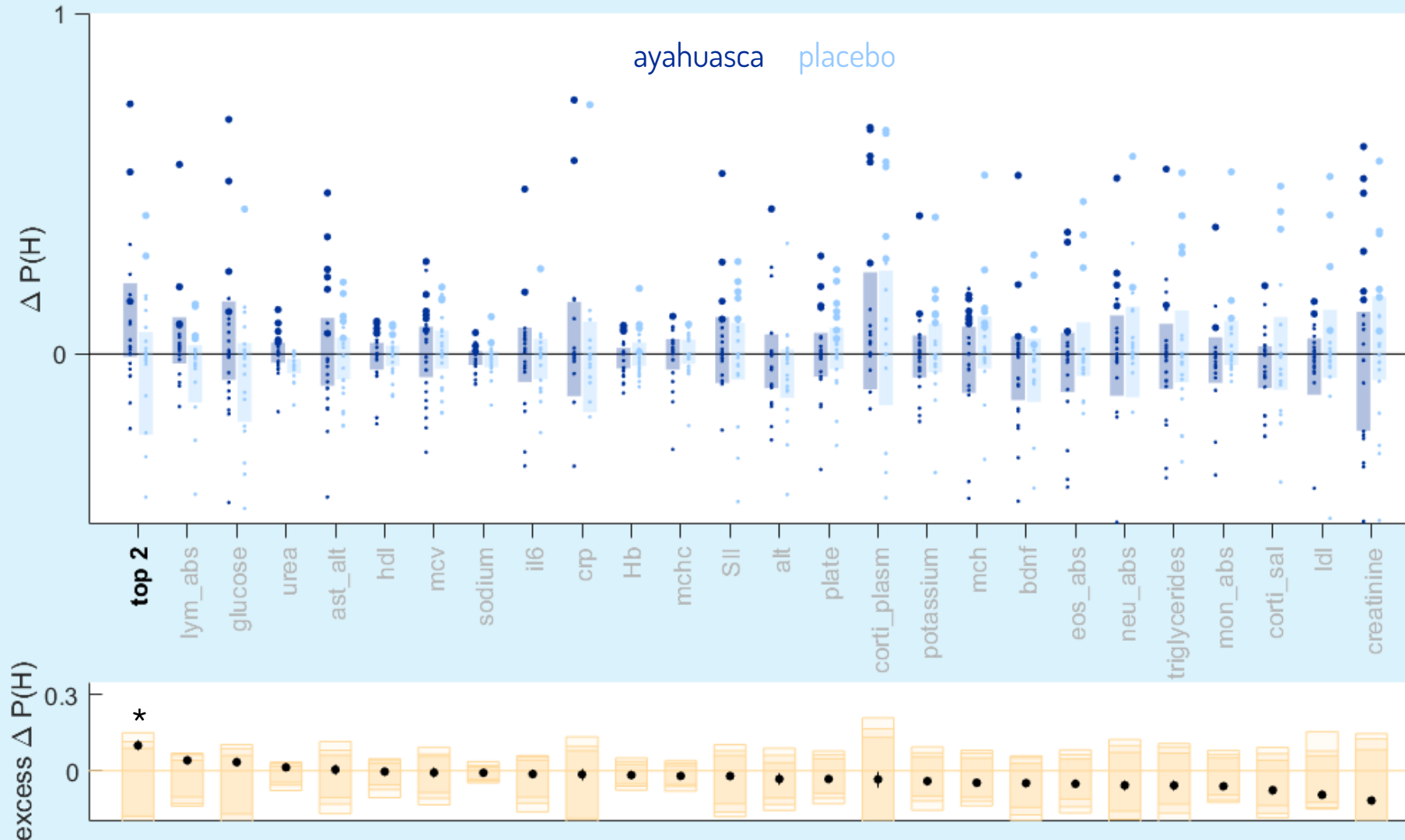
- Markers are ranked by:
 - how much aya treatment restores it towards baseline healthy
 - how much more aya restores it compared to placebo
- markers that most separate H/D are **not** the ones most restored.
- only **CRP** shows significant restoration by itself
- **top 2 markers combined** shows more significant restoration

Building an optimal combination of **restoration** markers

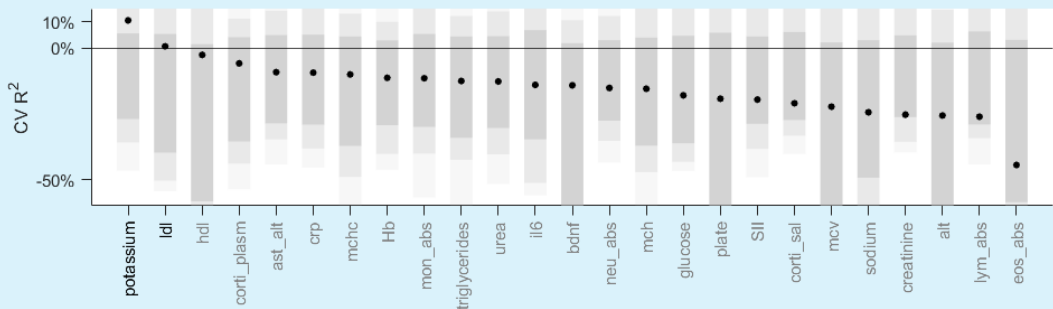
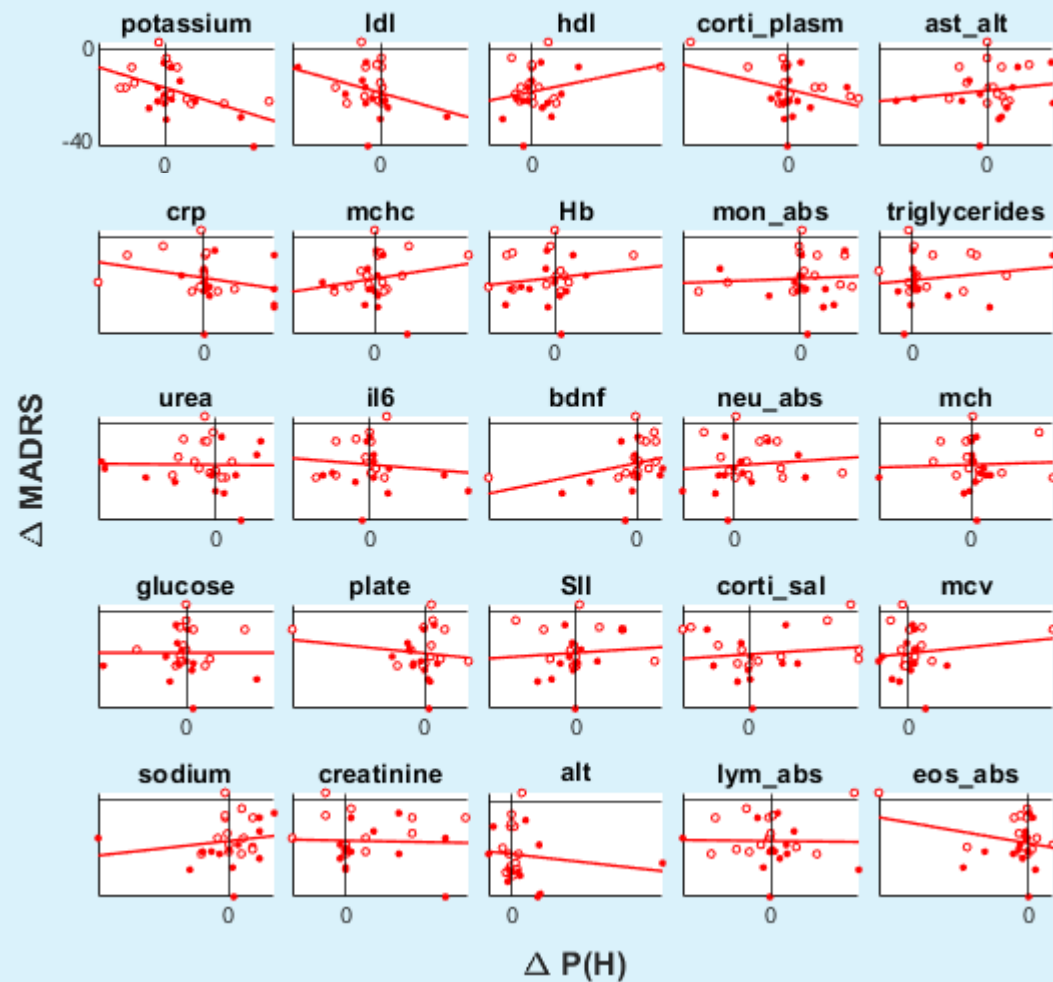


- Combined restoration again first rises (**underfitting**), then falls (**overfitting**).
- Combining **CRP** and **creatinine** is good enough

3. Are markers of the **healthy** restored *even more* after treatment?



- Psychedelics for the 'betterment of well people'
- $\Delta P(H)$ can show if the markers in the healthy shift **even closer to healthy values**.
- **No individual marker** showed such significant restoration.
- Combining the top 2: only marginally significant restoration.

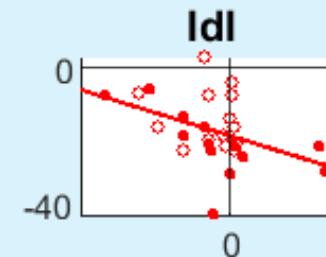
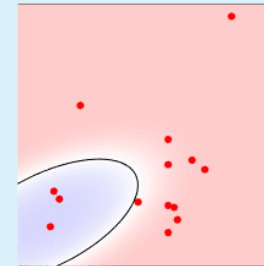
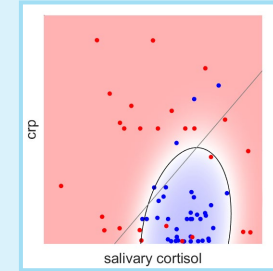


4. Which marker restorations correlate with feeling better?

- Plot against $\Delta P(H)$ of biomarker, instead of raw difference.
- We keep placebo patients.
- We want three criteria:
 - negative slope
 - cross-validated $R^2 > 0$
 - $p < 0.05$
- Only **potassium** and **LDL** shows a moderate effect.

Conclusions

- **Salivary cortisol and CRP** together best separate baseline healthy from depressed, by 84%.
- In the depressed, **CRP & creatinine** were restored most towards healthy values after ayahuasca treatment, compared to placebo.
- In the healthy, **no markers** were restored much after treatment.
- Restorations in **Potassium and LDL** are moderately correlated with improvements in depression.



Thanks



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